Fear Spikes and Next-Week Returns: Does an Elevated VIX Predict Higher US Equity Returns?

SIQuant Research[1]

[1]Hypothesis × Quant

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Abstract

Hypothesis: H_0: the 60-day VIX z-score has no predictive power for the S&P 500 return over the next five sessions; H_1: the slope is positive
Data: 2,450 daily observations, 2017-01-19 to 2026-10-16; 184 sessions with $z > 2$
Method: OLS with Newey–West HAC errors (10 lags) and a 2,000-draw circular-shift placebo; the decision needs both one-sided $p < 0.05$
Results: $\beta = -0.00023$ (HAC $t = -0.22$, $p = 0.5858$; placebo $p = 0.6067$), $R^2 = 0.0001$; mean next-week return 0.10% after $z > 2$ versus 0.26% otherwise.
Research decision: **REJECTED**

1 Design

The VIX z-score is computed from the 60 closes ending on day t and the outcome is the log return from the close of t to the close of $t + 5$, so predictor and outcome do not overlap [Whaley, 2000]. The five-day overlap between neighbouring outcomes is handled with HAC errors (10 lags) and a circular-shift placebo.

2 Results

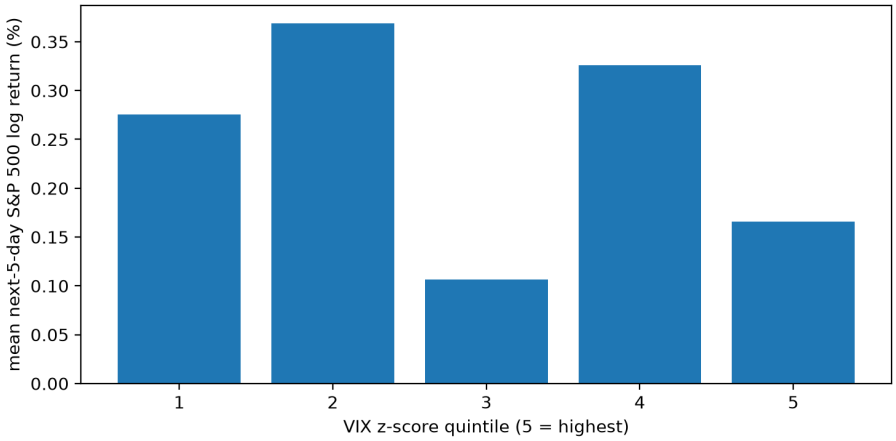


Figure 1: Mean next-week S&P 500 return by quintile of the VIX z-score.

3 Limitations

The sample contains few independent volatility spikes (2018, 2020, 2022, 2025), so the effective sample is small. No transaction costs or timing frictions are modelled and the result is not a trading rule.

References

Robert E Whaley. The investor fear gauge. *The Journal of Portfolio Management*, 26(3):12–17, 2000.